Assignment 1.1.

Introduction to Data Cleaning

Objectives:

- Analyze and interpret the structure and content of raw datasets.
- Apply foundational data preprocessing techniques, such as handling missing values and data type conversions.
- Construct a basic data cleaning pipeline to process real-world data efficiently and with reproducibility.

Procedures

Download the dataset for this activity here

1. Load the dataset then show the first 5 and last 5 datapoints

```
import pandas as pd

df = pd.read_csv("/content/drive/MyDrive/(1st Semester AY 2025 to 2026) CPE 031/dirty_cafe_sales.csv")
```

```
df.head()
```

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
2	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
3	TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
4	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11

```
df.tail()
```

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
9995	TXN_7672686	Coffee	2	2.0	4.0	NaN	UNKNOWN	2023-08-30
9996	TXN_9659401	NaN	3	NaN	3.0	Digital Wallet	NaN	2023-06-02
9997	TXN_5255387	Coffee	4	2.0	8.0	Digital Wallet	NaN	2023-03-02
9998	TXN_7695629	Cookie	3	NaN	3.0	Digital Wallet	NaN	2023-12-02
9999	TXN_6170729	Sandwich	3	4.0	12.0	Cash	In-store	2023-11-07

QUESTION: What Column has numerical values?

Enter your answer here: The dataset has three columns with numerical values: Quantity, Price Per Unit, and Total Spent. Quantity contains whole numbers showing how many items were bought, while Price Per Unit and Total Spent contain decimal values representing costs

2. Check if there are any missing values in the dataset

```
df.describe()
```

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
count	10000	9667	9862	9821	9827	7421	6735	9841
unique	10000	10	7	8	19	5	4	367
top	TXN_9226047	Juice	5	3.0	6.0	Digital Wallet	Takeaway	UNKNOWN
freq	1	1171	2013	2429	979	2291	3022	159

```
df.isna().sum()
```

	0
Transaction ID	0
Item	333
Quantity	138
Price Per Unit	179
Total Spent	173
Payment Method	2579
Location	3265
Transaction Date	159

dtype: int64

QUESTION: How many columns had missing values? Which column had the highest number of missing values?

Enter your answer here: Based on the dataset, the location column has the highest number of missing values.

3. Remove unneccessary columns and make sure numerical columns only contain numbers.

```
df.columns
```

```
Index(['Transaction ID', 'Item', 'Quantity', 'Price Per Unit', 'Total Spent',
       'Payment Method', 'Location', 'Transaction Date'],
      dtype='object')
```

```
df = df.drop(columns=['Transaction ID', 'Payment Method', 'Location', 'Transaction Date'])

df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2	2.0	4.0
1	Cake	4	3.0	12.0
2	Cookie	4	1.0	ERROR
3	Salad	2	5.0	10.0
4	Coffee	2	2.0	4.0

```
df.dtypes ## Checking the datatype for each column
```

	0
Item	object
Quantity	object
Price Per Unit	object
Total Spent	object

dtype: object

```
df["Quantity"] = pd.to_numeric(df["Quantity"], errors='coerce')
df["Price Per Unit"] = pd.to_numeric(df["Price Per Unit"], errors='coerce')
df["Total Spent"] = pd.to_numeric(df["Total Spent"], errors='coerce')

df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2.0	2.0	4.0
1	Cake	4.0	3.0	12.0
2	Cookie	4.0	1.0	NaN
3	Salad	2.0	5.0	10.0
4	Coffee	2.0	2.0	4.0

```
df.dtypes ## Checking the datatype for each column
```

	0
Item	object
Quantity	float64
Price Per Unit	float64
Total Spent	float64

dtype: object

QUESTION: What happened to the object columns?

Enter your answer here: The three numeric columns (Quantity, Price Per Unit, Total Spent) changed from object to float64, with invalid values like "ERROR" turned into NaN. The Item column stayed as object since it's text.

4. Remove missing data to ensure all that the final dataset is clean

```
df.isna().sum()
```

	0
Item	333
Quantity	479
Price Per Unit	533
Total Spent	502

dtype: int64

```
df = df.dropna()
```

```
df.isna().sum()
```

	0
Item	0
Quantity	0
Price Per Unit	0
Total Spent	0

dtype: int64

```
df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2.0	2.0	4.0
1	Cake	4.0	3.0	12.0
3	Salad	2.0	5.0	10.0
4	Coffee	2.0	2.0	4.0
5	Smoothie	5.0	4.0	20.0

EXPLAIN the process of data cleaning done in the procedure.

Enter your answer here: The data cleaning process involved a few key steps. First, unnecessary columns like Transaction ID, Payment Method, Location, and Transaction Date were removed since they weren't needed for the analysis. Next, the numeric columns (Quantity, Price Per Unit, and Total Spent) were converted using pd.to_numeric() with errors='coerce', which turned invalid entries like "ERROR" into NaN (missing values). Finally, df.dropna() was used to remove all rows containing missing values, leaving a clean dataset where every remaining row has complete, valid numeric data in all columns.

Supplemetary Activities

Download the dataset for supplementary activity here

1. Load the dataset and show the first 5 and last 5 datapoints
2. Check if there are any missing values in the dataset.
3. The df must only have Name, Subsribers, Category, and Country. Remove all other columns.
4. Convert Subscribers to type **int**
5. Ensure that there are no rows with missing values. (UNKNOWN is counted as a missing value)

```
import pandas as pd

df = pd.read_csv("youtube_subscribers_data.csv")
```

1. Load the dataset and show the first 5 and last 5 datapoints

```
df.head()
```

	Name	Brand channel	Subscribers (millions)	Primary language	Category	Country
0	MrBeast	No	335	English	Entertainment	United States
1	T-Series	UNKNOWN	280	Hindi	Music	India
2	Cocomelon - Nursery Rhymes	Yes	UNKNOWN	UNKNOWN	Education	United States
3	SET India	Yes	UNKNOWN	Hindi	Entertainment	India
4	Vlad and Niki	No	129	English	Entertainment	Russia

Next steps: Generate code with df | New interactive sheet

```
df.tail()
```

	Name	Brand channel	Subscribers (millions)	Primary language	Category	Country
45	ARY Digital HD	Yes	59.7	Urdu	Entertainment	Pakistan
46	Fede Vigevani	No	59.4	UNKNOWN	Entertainment	Mexico
47	LooLoo Kids	UNKNOWN	59.3	English	Music	Romania
48	BillionSurpriseToys	Yes	UNKNOWN	English	Entertainment	United States
49	Marshmello	No	57.5	English	Music	United States

2. Check if there are any missing values in the dataset.

```
df.describe()
```

	Name	Brand channel	Subscribers (millions)	Primary language	Category	Country
count	50	50	50	50	50	50
unique	50	3	42	12	11	16
top	MrBeast	Yes	UNKNOWN	English	Entertainment	India
freq	1	26	7	19	20	17

```
df.isna().sum()
```

	0
Name	0
Brand channel	0
Subscribers (millions)	0
Primary language	0
Category	0
Country	0

dtype: int64

3. The df must only have Name, Subsribers, Category, and Country. Remove all other columns.

```
df = df[['Name', 'Subscribers (millions)', 'Category', 'Country']]
df.head()
```

	Name	Subscribers (millions)	Category	Country
0	MrBeast	335	Entertainment	United States
1	T-Series	280	Music	India
2	Cocomelon - Nursery Rhymes	UNKNOWN	Education	United States
3	SET India	UNKNOWN	Entertainment	India
4	Vlad and Niki	129	Entertainment	Russia

Next steps: Generate code with df | New interactive sheet

```
df.dtypes
```

	0
Name	object
Subscribers (millions)	object
Category	object
Country	object

dtype: object

4. Convert Subscribers to type int

```
df["Subscribers (millions)"] = pd.to_numeric(df["Subscribers (millions)"], errors='coerce')
df["Subscribers (millions)"] = df["Subscribers (millions)"].astype(int)
df.head()
```

	Name	Subscribers (millions)	Category	Country
0	MrBeast	335	Entertainment	United States
1	T-Series	280	Music	India
4	Vlad and Niki	129	Entertainment	Russia
5	Kids Diana Show	128	Entertainment	United States Ukraine
6	Like Nastya	123	Entertainment	United States Ukraine

Next steps: Generate code with df | New interactive sheet

```
df.dtypes
```

	0
Name	object
Subscribers (millions)	int64
Category	object
Country	object

dtype: object

5. Ensure that there are no rows with missing values. (UNKNOWN is counted as a missing value)

```
df.replace("UNKNOWN", pd.NA)
df.dropna()
```

	Name	Subscribers (millions)	Category	Country
0	MrBeast	335	Entertainment	United States
1	T-Series	280	Music	India
4	Vlad and Niki	129	Entertainment	Russia
5	Kids Diana Show	128	Entertainment	United States Ukraine
6	Like Nastya	123	Entertainment	United States Ukraine
7	Zee Music	112	Music	India
8	PewDiePie	110	Entertainment	Sweden Japan
9	WWE	105	Sports	United States
11	Stokes Twins	100	Entertainment	United States
12	Sony SAB	97	Entertainment	India
13	Blackpink	95	Music	South Korea
14	ChuChu TV Nursery Rhymes & Kids Songs	94	Education	India
15	Zee TV	86	Entertainment	India
16	5-Minute Crafts	81	How-to	Cyprus[a]
17	BANGTANTV	79	Music	South Korea
18	Pinkfong	79	Education	South Korea
19	Colors TV	77	Entertainment	India
20	Hybe Labels	76	Music	South Korea
21	Justin Bieber	74	Music	Canada
22	T-Series Bhakti Sagar	72	Music	India
23	김프로KIMPRO	71	Entertainment	South Korea
24	UR · Cristiano	71	Entertainment/Sports	Portugal
25	A4	71	Entertainment	Belarus
26	Tips Official	70	Entertainment	India
27	Shemaroo Filmi Gaane	70	Music	India
28	Alan Chikin Chow	69	Entertainment	United States
29	ZAMZAM ELECTRONICS TRADING	68	Entertainment	United Arab Emirates
30	Aaj Tak	68	News	India
32	Canal KondZilla	67	Music	Brazil
33	El Reino Infantil	67	Music	Argentina
35	Wave Music	63	Music	India
36	Eminem Music	63	Music	United States
37	YRF	62	Music	India
38	Movieclips	62	Film	United States
39	Sony Music India	62	Music	India
41	KL BRO Biju Rithvik	62	Lifestyle Vlogs	India
42	Dude Perfect	60	Comedy	United States
43	Taylor Swift	60	Music	United States
44	Mark Rober	60	Education/Entertainment	United States
45	ARY Digital HD	59	Entertainment	Pakistan
46	Fede Vigevani	59	Entertainment	Mexico
47	LooLoo Kids	59	Music	Romania
49	Marshmello	57	Music	United States

Conclusion

In this activity, I learned how to clean and prepare data using the Pandas library. I practiced selecting only the needed columns, converting data types, and handling missing values by treating "UNKNOWN" as missing data and removing incomplete rows. This

activity showed me that data cleaning is an important step because accurate and consistent data leads to more reliable analysis and results.

Good job on finishing the activity! Kindly submit your accomplished google colab link and a PDF file of this activity.